

INDEPENDENT RESEARCH REPORT

Part IV: Documented Iatrogenic Harms by Drug Class

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1 Part IV: Documented Iatrogenic Harms by Drug Class

1.1 Overview and Methodology

Part III of this report demonstrated that the antipsychotic evidence base fails the evidentiary standard for justified coercive treatment on Criteria A, B, and C: the mechanism of action remains unverified, the primary outcome measure is contaminated by sedation, and long-term functional outcomes are worse under maintained antipsychotic treatment than under guided dose reduction. This Part addresses Criterion D: whether the benefits of antipsychotics demonstrably and substantially outweigh their harms at the population level.

The conclusion, demonstrated below, is that they do not.

The framework for harm assessment uses standard pharmacovigilance metrics where available. The *Number Needed to Harm* (NNH) is the number of patients who must receive the drug for one additional patient to experience the specified harm relative to what would have occurred without treatment. An NNH of 3 means that for every three patients treated, one experiences that harm who would not have otherwise. The NNH is most meaningful when placed alongside the corresponding Number Needed to Treat (NNT) for the claimed benefit: if NNT and NNH are similar, benefits and harms are roughly balanced; if NNH is substantially lower than NNT, harm predominates.

The harm profiles below are drawn from three tiers of evidence: regulatory prescribing information (which carries the highest standard for adverse effect documentation because manufacturers are legally required to disclose); peer-reviewed systematic reviews and meta-analyses; and the CATIE and EUFEST independent trial datasets reviewed in Part III.

This Part makes one preliminary acknowledgement. Antipsychotics produce real pharmacological effects. For some patients in acute psychotic crisis, they reduce behavioural disturbance and distress on a short-term basis. The argument here is not that these effects are zero. The argument is that the documented harms are of a magnitude, prevalence, and permanence that cannot be justified in a system that

administers these drugs coercively to patients who have not consented and cannot meaningfully refuse.

The orienting finding comes from De Hert et al. (2011), published in *World Psychiatry*:

De Hert et al. (2011), *World Psychiatry*

“Patients with severe mental disorders have a 2–3 times higher risk of dying from cardiovascular and metabolic disorders compared with the general population. Evidence suggests that antipsychotic drugs *further increase* the already elevated cardiovascular and metabolic risk in this population.”

This finding establishes the policy context: antipsychotics are being administered coercively to a population already at elevated risk of the harms these drugs most reliably produce. The drugs are amplifying, not reducing, the physical health vulnerability of the people they are meant to help.

1.2 First-Generation (Typical) Antipsychotics

1.2.1 *Class Overview and Historical Context*

First-generation antipsychotics (FGAs) were introduced beginning with chlorpromazine in 1952 and haloperidol in 1958. They act primarily through high-affinity dopamine D2 receptor blockade. They remain in common use in Australia, particularly in acute psychiatric emergency settings, in depot (long-acting injectable) formulations administered under Community Treatment Orders, and in facilities serving older populations.

The marketing of second-generation antipsychotics from the 1990s created a narrative in which FGAs were retrospectively recast as primitive and dangerous, to be replaced by the safer, more effective atypicals. As the independent CATIE and EUFEST trial data established, this narrative was substantially industry-constructed. The first-generation drug perphenazine performed comparably to the second-generation drugs in CATIE on the primary effectiveness outcome; the genuine difference between drug generations is not in therapeutic efficacy but in the *type* of harm profile, with FGAs producing more neurological harms and the newer drugs more metabolic ones.

1.2.2 *Chlorpromazine (Largactil)*

Chlorpromazine is the original antipsychotic, the prototype against which all subsequent drugs are compared, and the drug whose discovery inaugurated the modern era of biological psychiatry. Its harm profile, established across seven decades of use in tens of millions of patients, is among the most thoroughly documented in pharmacology:

- **Extrapyramidal side effects (EPS):** Parkinsonism (bradykinesia, shuffling

gait, masked facies, tremor, rigidity), acute dystonic reactions (sustained, painful muscle contractions, often of the neck, jaw, or eyes), and akathisia (see Section 1.11 below). NNH for EPS with FGAs versus placebo is approximately 3–5 across meta-analyses;

- **Sedation:** Profound histamine H1-receptor-mediated sedation is typically the most prominent patient-perceived effect at treatment initiation. This sedation is pharmacologically non-specific — it is a side effect identical to that produced by antihistamines — and is not evidence of antipsychotic action. As discussed in Part III, this sedation substantially inflates the apparent efficacy of antipsychotics on PANSS-based outcome measures;
- **Anticholinergic effects:** Dry mouth, constipation (which can progress to paralytic ileus at high doses), urinary retention, tachycardia, blurred vision, and cognitive impairment including delirium in elderly patients;
- **Tardive dyskinesia (TD):** Cumulative incidence with long-term FGA use is approximately 20–30% at five years, rising to over 50% with very long-term exposure in some cohorts (Carbon et al., 2018). TD is detailed in Section 1.10;
- **QTc prolongation:** Cardiac conduction delay, particularly at higher doses and in combination with other QT-prolonging agents. QTc prolongation beyond 500 ms is associated with torsades de pointes, a potentially fatal ventricular arrhythmia;
- **Photosensitivity:** Severe sunburn reactions at solar radiation doses that would be innocuous without the drug. Clinically significant in the Australian context given sun exposure levels;
- **Pigmentary retinopathy and lenticular opacities:** Documented at higher doses and with prolonged use, causing progressive visual impairment;
- **Neuroleptic malignant syndrome (NMS):** A rare but potentially fatal idiosyncratic adverse reaction characterised by hyperthermia, severe muscle rigidity (lead-pipe rigidity), altered consciousness, and autonomic instability. Untreated mortality is approximately 10–20%. Immediate drug cessation and intensive medical management are required; however, for a patient on a depot CTO, drug cessation is not immediately achievable.

1.2.3 *Haloperidol (Haldol, Serenace)*

Haloperidol is the most commonly used high-potency FGA and remains a first-line acute management agent in many Australian emergency departments and inpatient units, where it is frequently administered by intramuscular injection for rapid tranquillisation — that is, as a means of pharmacological restraint, without the patient’s consent, in the context of the acute coercive cascade documented in Part II.

Its harm profile relative to chlorpromazine shows reduced sedation and metabolic burden but substantially higher rates of neurological harm:

- **EPS at highest rates of any major antipsychotic:** In CATIE, the first-generation comparator (perphenazine) produced significantly higher rates of EPS than any of the four second-generation comparators. Haloperidol, which is more potent than perphenazine, produces higher EPS rates still. The NNH for clinically significant EPS with haloperidol versus placebo across the dose range used in practice is approximately 3–5;
- **Acute dystonia:** Particularly common in young men and with rapid dose escalation. Oculogyric crises (involuntary upward eye deviation), torticollis, and opisthotonus (painful arching of the back) are documented. These are acute medical emergencies;
- **Akathisia at high rates:** NNH for akathisia with haloperidol versus placebo is approximately 5–8 in short-term trials;
- **Tardive dyskinesia:** Among the highest rates of any antipsychotic with long-term use;
- **Haloperidol decanoate (depot):** Widely used under CTOs in Australia. The depot formulation is injected intramuscularly and releases drug continuously for two to four weeks. Once administered, it cannot be removed. If the patient develops NMS, severe akathisia, or any other serious adverse reaction following depot injection, the drug continues to be released for weeks regardless. This makes genuinely revocable informed consent a structural impossibility for this formulation. It is, by definition, an irreversible coercive act.

1.3 Summary Harm Profiles: All Major Antipsychotics

Table 1 provides a comparative overview of the relative harm profiles of the major antipsychotics in current Australian use. The ratings reflect the weight of evidence across systematic reviews, meta-analyses, and the CATIE and EUFEST independent trials.

Table 1: Comparative harm profiles: major antipsychotics in Australian use

Drug	Weight Gain	Diabetes	Tardive Dyskinesia	EPS	Sedation	Prolactin
Clozapine	+++++	++++	+	+	++++	+
Olanzapine	+++++	+++	++	+	+++	+
Quetiapine	+++	++	+	+	+++	+
Chlorpromazine	++	++	+++	+++	++++	+++
Risperidone	++	++	++	++	+	++++
Haloperidol	+	+	++++	++++	+	+++
Aripiprazole	+	+	+	+	+	↓

Rating scale: + minimal; ++ low; +++ moderate; ++++ high; +++++ very high; ↓ below baseline (prolactin-lowering). Compiled from De Hert et al. (2011), Carbon et al. (2018), CATIE (Lieberman et al., 2005), and individual drug regulatory prescribing information. Relative profiles only; not a substitute for current prescribing information. Aripiprazole is a D2 partial agonist; its lower harm ratings in some categories reflect a different mechanism, not absence of harm (see compulsive behaviour risk below).

1.4 Second-Generation Antipsychotics: Overview

Second-generation antipsychotics (SGAs) were introduced from the early 1990s with aggressive pharmaceutical marketing presenting them as both more effective and safer than FGAs. This claim was driven almost entirely by industry-sponsored head-to-head trials, which as Heres et al. (2006) documented, reported the sponsor’s drug as superior in 90% of cases — a mathematically impossible result in a neutrally conducted literature.

The independent CATIE and EUFEST trial data, reviewed in Part III, found no significant efficacy advantage for second-generation drugs over perphenazine (an FGA). The genuine difference between generations is not therapeutic superiority but a *shift* in the harm profile: SGAs produce less neurological harm (lower rates of EPS and TD) but substantially greater metabolic harm (weight gain, diabetes, dyslipidaemia). For patients who would develop tardive dyskinesia on an FGA, this is a meaningful improvement. For patients who develop diabetes, cardiovascular disease, or premature death from the SGA’s metabolic burden, the improvement is illusory.

The drug-by-drug profiles that follow detail the specific harm burden of each major SGA in Australian use.

1.5 Olanzapine (Zyprexa)

Olanzapine was introduced in 1996 by Eli Lilly and rapidly became one of the most widely prescribed antipsychotics globally. In Australia, it remains among the most commonly prescribed drugs in this class. It was marketed as a breakthrough on both efficacy and safety grounds. The post-marketing evidence, particularly from CATIE, told a different story.

1.5.1 *Weight Gain and Metabolic Harm*

Olanzapine produces the most severe metabolic effects of any currently registered antipsychotic in common Australian use:

- **Weight gain:** Mean weight gain in short-term (10-week) placebo-controlled trials is 4–5 kg. In CATIE, patients on olanzapine gained approximately 0.9 kg per month throughout the 18-month study period, with no plateau — suggesting weight gain continues indefinitely with ongoing treatment. NNH for clinically significant weight gain ($\geq 7\%$ of body weight) at 10 weeks is approximately 3: for every three patients treated, one will experience this degree of weight gain who would not have without the drug;
- **New-onset type 2 diabetes mellitus:** Documented across both epidemiological cohort studies and clinical trial data. De Hert et al. estimated NNH for diabetes over a three-year exposure period at approximately 25–50, varying substantially by baseline metabolic risk. In populations with pre-existing obesity, impaired glucose tolerance, or family history of diabetes — which describes a significant proportion of people with schizophrenia — the NNH is substantially lower;
- **Dyslipidaemia:** Significant elevation of total cholesterol, LDL cholesterol, and triglycerides. In CATIE, olanzapine produced significantly greater increases in triglycerides and total cholesterol than all comparator drugs. The cardiovascular risk implications compound across decades of use;
- **Non-alcoholic fatty liver disease (NAFLD):** Documented at higher rates than comparator antipsychotics, attributable to the combination of weight gain, insulin resistance, and possible direct hepatic effects of the drug.

The metabolic burden of olanzapine is not merely uncomfortable. Obesity, type 2 diabetes, and dyslipidaemia are the leading drivers of cardiovascular disease mortality. They are the pathways through which the 14–17-year life expectancy gap documented in Australian data (Lawrence et al., 2013) is generated. Olanzapine — the drug with the best discontinuation profile in CATIE — produces the greatest acceleration of these mortality pathways.

1.5.2 Corporate Suppression of Metabolic Evidence

The metabolic harm of olanzapine was not unknown to its manufacturer. Internal Eli Lilly documents, disclosed through United States litigation, showed that the company maintained internal records documenting the drug's metabolic effects — including weight gain, elevated glucose, and diabetic events — years before this information was adequately reflected in the prescribing information provided to clinicians and patients. A corporate strategy document from the late 1990s explicitly discussed how to minimise the commercial impact of the weight gain data.

In 2009, Eli Lilly entered a federal guilty plea and paid approximately \$1.4 billion USD to resolve criminal and civil allegations related to the illegal off-label marketing of Zyprexa. This is the largest criminal fine paid by an individual company in United States history at that time.

This is not an incidental footnote. The drug that is coercively administered to tens of thousands of Australians each year, whose metabolic harm profile is now well-documented, was the subject of deliberate corporate evidence suppression for at least a decade. The prescribing decisions made during that decade, and the clinical guidelines that were established during it, were made on the basis of a deliberately distorted evidence base.

1.5.3 Drug-Induced Secondary Negative Symptoms

Olanzapine's profound sedation produces a clinical syndrome that closely resembles the negative symptoms of schizophrenia — reduced motivation, blunted affect, poverty of speech, social withdrawal, and cognitive slowing. These drug-induced effects are frequently misattributed by clinicians to the patient's underlying illness rather than to the treatment, leading to:

- Dose escalation in response to apparent negative symptoms that are themselves drug-caused;
- Attribution of functional impairment to the illness rather than to the treatment, shielding the drug from accountability;
- Reduced family and community expectations of recovery, because the drug-induced syndrome reinforces the narrative of permanent disability.

This mechanism has never been systematically studied in Australian populations on CTOs, where dose is determined by the treating team rather than patient preference, and where adverse effect reports may be attributed to non-compliance rather than to drug toxicity.

1.6 Risperidone and Paliperidone (Risperdal, Invega)

Risperidone was introduced in 1994 and paliperidone (its active metabolite) followed in the 2000s. Both remain extensively prescribed in Australia, including in depot (long-acting injectable) formulations: Risperdal Consta (risperidone, two-weekly), Invega Sustenna (paliperidone, monthly), and Invega Trinza (paliperidone, three-monthly).

1.6.1 *Hyperprolactinaemia: Prevalence and Consequences*

Risperidone and paliperidone produce the most pronounced prolactin elevation of any commonly used antipsychotic. Dopamine normally inhibits prolactin release from the pituitary lactotroph cells via D2 receptor activation; D2 blockade removes this inhibition, producing continuous prolactin hypersecretion.

Prolactin elevation occurs in virtually all patients treated with risperidone or paliperidone at therapeutic doses. The clinical consequences are extensive, often unaddressed, and frequently unreported:

- **Sexual dysfunction:** Reduced libido (reported by 30–50% of patients), erectile dysfunction in men, anorgasmia in both sexes, and ejaculatory dysfunction. These effects directly impair quality of life and intimate relationships. They are frequently not disclosed by patients who feel shame or who assume their treating clinician will not take them seriously;
- **Menstrual irregularity and amenorrhoea:** Elevated prolactin suppresses the hypothalamic-pituitary-gonadal axis, producing menstrual irregularity and amenorrhoea in women. This may be misinterpreted as pregnancy or as an unrelated gynaecological problem;
- **Galactorrhoea:** Inappropriate breast milk production occurs in both women and men (gynaecomastia with galactorrhoea in the latter). This is distressing, stigmatising, and directly caused by the drug;
- **Reduced bone mineral density:** Chronic hypo-oestrogenism or hypo-androgenism resulting from prolactin-mediated gonadal suppression progressively reduces bone mineral density, increasing the risk of osteoporosis and fragility fractures over years of treatment;
- **Cancer risk:** Elevated prolactin promotes proliferation of prolactin-receptor-positive breast tissue. Solmi et al. (2023) conducted a systematic review and meta-analysis of 14 observational studies involving approximately 500,000 patients, finding a statistically significant association between antipsychotic use and breast cancer risk (relative risk approximately 1.3–1.7), with the association strongest for prolactin-elevating antipsychotics. The biological mechanism is established; the dose-response relationship is consistent with causation. The NNH for prolactin-related cancer over long-term exposure has not been precisely quantified but is

not negligible.

1.6.2 Extrapyramidal Side Effects

Risperidone produces higher rates of extrapyramidal side effects than olanzapine or quetiapine, though lower than haloperidol at equivalent doses. However, in clinical practice, risperidone is frequently prescribed at doses above 6 mg/day — at which point EPS rates approach those of first-generation drugs. This dose escalation is common under CTOs, where the treating team responds to perceived “breakthrough” symptoms by increasing dose rather than reconsidering the treatment approach.

1.6.3 Depot Formulations and the Impossibility of Revocable Consent

The three-monthly paliperidone palmitate injection (Invega Trinza) is specifically marketed for patients with “poor medication compliance” — a population that substantially overlaps with CTO patients who have been legally compelled to accept treatment they do not want. Once administered, the drug is released continuously from the depot for approximately three months. A patient who develops severe adverse effects within days of the injection — including worsening akathisia with suicidal ideation, NMS, or severe metabolic deterioration — cannot have the drug removed. They must endure its continued release for the full three-month period.

This is not a manageable clinical inconvenience. It is a fundamental violation of the principle that medical treatment must be revocable. The use of three-monthly depot formulations under CTOs represents the most extreme version of the coercive cascade: legally compelled, pharmacologically irrevocable treatment for a minimum of three months per dose.

1.7 Clozapine (Clopine, Clozaril)

Clozapine is the only antipsychotic with genuinely demonstrated superiority for treatment-resistant schizophrenia, defined as illness that has not responded to two adequate trials of other antipsychotics. This clinical utility is real, is acknowledged in this report, and must not be dismissed. For patients with genuine treatment-resistant illness who have had the risks explained and have made an informed choice to try clozapine, the drug may provide benefit that other treatments do not.

However, the harm profile of clozapine is the most severe of any antipsychotic in current use, and its administration in settings where the monitoring requirements cannot be consistently met — including community settings under CTOs — raises serious safety concerns.

1.7.1 Agranulocytosis

Clozapine causes agranulocytosis — a severe, potentially fatal reduction in neutrophil count that leaves the patient acutely immunocompromised — in approximately 0.5–2% of patients. Untreated agranulocytosis results in overwhelming bacterial sepsis and death. The risk is highest in the first six months of treatment but persists throughout.

Mandatory white blood cell monitoring is a condition of clozapine’s regulatory approval in Australia: weekly for the first 18 weeks, then fortnightly. This monitoring requirement is not optional clinical practice; it is a legal condition of prescribing. However:

- In community settings and for patients with chaotic lives, consistent fortnightly blood testing is frequently not achieved;
- Patients on CTOs who are required to attend for depot medication may not also have access to convenient blood testing facilities;
- There are documented cases in Australia of clozapine being inaccessible to patients who could benefit from it precisely because the monitoring infrastructure cannot be reliably provided.

The monitoring requirement also creates a pharmacological access inequality: clozapine is most accessible to patients with stable living situations, reliable transport, and good physical health — patients who, by definition, are not those at highest need.

1.7.2 Myocarditis and Cardiomyopathy

Clozapine is associated with myocarditis (inflammation of the myocardium) in approximately 2–5% of patients, with the risk concentrated in the first month of treatment. Clozapine-associated myocarditis is potentially fatal: if unrecognised and the drug not discontinued, the inflammatory process can progress to cardiomyopathy and cardiac failure.

Ronaldson et al. (2011) documented a 2.8% rate of clozapine-induced myocarditis in a large Australian cohort — one of the most rigorous population-based estimates available. Critically, this Australian study found mortality rates among those who developed myocarditis to be approximately 30% if diagnosis was delayed or the drug not promptly discontinued. Australia has the highest documented rate of clozapine-induced myocarditis in the world, attributed in part to the faster titration protocols used in some Australian centres.

The combination of agranulocytosis and myocarditis risk means that clozapine initiation requires a level of medical surveillance — frequent blood testing, baseline ECG, serial troponin measurements, regular clinical review — that is appropriate in well-resourced inpatient or specialist outpatient settings but difficult to guarantee in

community settings. Prescribing clozapine under a CTO, in circumstances where the monitoring requirements cannot be assured, represents a serious patient safety risk.

1.7.3 Metabolic Effects

Clozapine produces weight gain and metabolic effects comparable to olanzapine in severity — among the worst of all antipsychotics. The combination of severe metabolic toxicity with the cardiovascular risks of myocarditis and cardiomyopathy creates a compounding physical health burden that, without systematic monitoring and management, will substantially contribute to the life expectancy gap documented in Lawrence et al. (2013).

1.8 Quetiapine (Seroquel)

Quetiapine was introduced in 1997 and is among the most widely prescribed psychotropic drugs in Australia across both psychiatric and general practice settings. Its prescribing pattern is unusual: while it is a registered antipsychotic approved for schizophrenia and bipolar disorder, a substantial proportion of quetiapine prescriptions in Australia and internationally are for off-label indications — primarily insomnia, anxiety, and unipolar depression — at doses (25–100 mg) substantially below the antipsychotic range.

1.8.1 Metabolic Profile

At antipsychotic doses (300–800 mg/day), quetiapine produces significant weight gain (less than olanzapine and clozapine but clinically meaningful) and dyslipidaemia. In CATIE, quetiapine produced the shortest median time to discontinuation of all five drugs in the trial — just 4.6 months — primarily driven by sedation and inefficacy rather than severe metabolic events. This result is consistent with a drug that produces significant burden without proportionate benefit.

1.8.2 The Off-Label Prescribing Problem

The widespread off-label prescribing of low-dose quetiapine for insomnia represents a distinct and underappreciated public health problem. At doses of 12.5–100 mg, quetiapine produces substantial histamine H1 receptor blockade — the mechanism responsible for its sedating effects — with relatively limited D2 blockade. It functions, at these doses, as an expensive, poorly characterised antihistamine.

However, low-dose quetiapine is not benign:

- **Metabolic effects persist at low doses:** Weight gain, glucose dysregulation, and dyslipidaemia are documented even at sub-antipsychotic doses in insomnia prescribing;

- **Pharmacological dependency:** Chronic H1 receptor blockade produces up-regulation of H1 receptors. When quetiapine is discontinued, the upregulated H1 system produces rebound insomnia that is worse than the original complaint, creating a dependency that makes the drug difficult to discontinue. Patients are rarely informed of this risk at initiation;
- **QTc prolongation at any dose:** QTc prolongation is a class effect that occurs at low doses, with the attendant risk of serious arrhythmia particularly in patients on other QT-prolonging agents;
- **Corporate misconduct:** AstraZeneca paid \$520 million USD in a 2010 United States settlement for illegal off-label marketing of Seroquel for uses including dementia and insomnia — the same pattern of deliberate commercial distortion of the evidence base documented for olanzapine/Eli Lilly.

1.9 Aripiprazole (Abilify)

Aripiprazole was approved in 2002 and represents a pharmacologically distinct approach: it is a partial agonist at D2 and D3 dopamine receptors and a partial agonist at serotonin 5-HT1A receptors, rather than a full antagonist at D2 like all other antipsychotics. This profile gives it a genuinely different harm profile — notably lower weight gain, lower prolactin elevation (it is mildly prolactin-lowering), and lower rates of most metabolic effects — but it introduces a distinct and severe adverse effect not seen with other antipsychotics.

1.9.1 *Impulse Control and Compulsive Behaviour: A Distinct and Serious Harm*

The partial D2 agonism of aripiprazole stimulates rather than blocks the dopamine reward system in the limbic and striatal circuits involved in motivation, reward, and habit formation. This stimulatory effect on reward pathways can, in susceptible individuals, produce a syndrome of pathological impulse dyscontrol.

Moore et al. (2014), published in *JAMA Internal Medicine*, reviewed 184 cases of aripiprazole-associated compulsive behaviour spontaneously reported to the US FDA adverse event reporting system. The behaviours documented included:

- **Pathological gambling:** The most commonly reported compulsive behaviour. Patients with no prior gambling history develop an inability to stop gambling, leading to loss of life savings, homes, businesses, and marriages;
- **Hypersexuality:** Compulsive sexual behaviour, including behaviour that creates legal and relationship consequences;
- **Compulsive shopping and spending:** Financial ruin from inability to control purchasing behaviour;

- **Binge eating:** Compulsive food intake inconsistent with any pre-existing eating disorder pattern.

The association is mechanistically plausible — the same pathway through which dopamine agonists (pramipexole, ropinirole) used in Parkinson’s disease cause impulse control disorders — and is drug-specific: Moore et al. found no similar signal for other antipsychotics in the same reporting database. The US FDA issued a black-box warning on aripiprazole for compulsive behaviour in 2016; the Australian Therapeutic Goods Administration issued a corresponding safety advisory.

Litigation in multiple jurisdictions has resulted in significant settlements for patients who suffered financial ruin, relationship breakdown, and in some documented cases, suicide attributable to aripiprazole-induced impulse dyscontrol.

The implications for coercive prescribing are severe. A patient administered aripiprazole under a Community Treatment Order, who then develops pathological gambling and loses their home, has been compelled by law to take a drug that destroyed their financial life. No monitoring mechanism exists to detect the early signs of impulse control disorder in CTO patients. No compensation framework exists. No systematic audit of this harm in the Australian CTO population has been conducted or published.

1.9.2 Akathisia

Despite its different pharmacological profile, aripiprazole produces some of the highest rates of akathisia of any second-generation antipsychotic, reported at 9–13% in registration trials. The partial agonist mechanism may produce akathisia through a different pathway than full antagonists, but the clinical syndrome is identical: distressing motor restlessness, inner agitation, and the associated suicide risk detailed in Section 1.11 below.

1.10 Cross-Cutting Harm I: Tardive Dyskinesia

1.10.1 Nature, Mechanism, and Clinical Presentation

Tardive dyskinesia (TD) is a movement disorder caused by chronic dopamine D2 receptor blockade and the resultant supersensitisation of the dopamine system described in Part III. It manifests as involuntary, repetitive, purposeless movements that the patient cannot suppress, are visible to others, and are directly caused by the drug.

The most common presentation involves the orofacial region: lip-smacking, tongue thrusting or rolling, chewing movements, puckering, and facial grimacing. Limb and trunk involvement also occurs: repetitive, writhing movements of the fingers, hands, and feet (athetoid movements) and rhythmic rocking, swaying, or twisting of the trunk.

In severe cases, TD impairs speech articulation, swallowing, and ambulation. The movements are uncontrollable, continuous, and visible in social interactions.

TD is neurologically disfiguring and profoundly socially stigmatising. In a social context that already carries enormous stigma for people with mental illness, the visible involuntary movements of tardive dyskinesia — movements that mark the person as different, as unwell, as somehow “other” — add a permanent visible label to a person’s experience of psychiatric treatment. This stigmatising dimension is rarely acknowledged in the clinical literature.

1.10.2 Prevalence: The Carbon et al. Meta-Analysis

Carbon et al. (2018), published in *World Psychiatry*, conducted the most comprehensive systematic review and meta-analysis of tardive dyskinesia prevalence to date, analysing data from 41 studies covering 57,563 patients:

Carbon et al. (2018), *World Psychiatry* — Tardive Dyskinesia Prevalence

- **Prevalent TD in patients on second-generation antipsychotics:** 20.7% (95% CI 19.3–22.2%);
- **Incident TD in randomised controlled trials** (shorter exposure periods): 7.2% for second-generation drugs vs. 7.8% for first-generation drugs — not significantly different, contradicting the marketing claim that SGAs are substantially safer for TD;
- **NNH for TD** with long-term first-generation antipsychotic use: approximately 5 at five years of exposure;
- **Risk factors for TD:** Older age, female sex, longer duration of treatment, higher cumulative dose, prior EPS, and African or Caribbean ancestry substantially increase risk. In high-risk subpopulations, the five-year NNH is considerably lower than 5;
- **Spontaneous remission rate:** Approximately one-third of TD cases remit after antipsychotic discontinuation; the remaining two-thirds persist indefinitely or permanently.

1.10.3 Translating Prevalence to Australian Population Estimates

Applying the 20.7% prevalence figure to the Australian antipsychotic prescribing population produces the following estimate. Approximately 300,000–350,000 Australians received antipsychotic prescriptions in recent years. Applying a conservative 15% TD prevalence to this population yields approximately 45,000–52,000 Australians with TD at any given time, caused directly by antipsychotic treatment.

Of these, approximately two-thirds — 30,000–35,000 people — have persistent or permanent TD that will not remit even if the drug is discontinued. These are people with a permanent, drug-caused, visible disability for which there is no compensation framework, no systematic monitoring program, and no acknowledgement in any Australian health policy document reviewed for this report.

1.10.4 The Irreversibility Problem and Compulsory Treatment

The irreversibility of TD in a substantial proportion of cases creates a specific ethical and legal problem in the context of compulsory treatment that has no equivalent in voluntary prescribing. When a drug is administered to a consenting patient who has been informed of the risks, the development of a permanent disability is a tragic outcome of an informed choice. When the same drug is administered under legal compulsion, without consent, and produces a permanent disability, the ethical weight of that outcome is categorically different.

No Australian Mental Health Act contains a mechanism for:

- Systematically monitoring TD rates in patients on CTOs;
- Reporting TD incidence attributable to compulsory treatment to any regulatory or oversight body;
- Providing compensation to patients who develop TD as a direct result of compulsory treatment;
- Mandating cessation of the causative drug when TD is detected, independent of the treating clinician's judgment.

TD caused by compulsory treatment is, in every jurisdiction in Australia, a harm for which the legal and healthcare system bears no formal accountability.

1.11 Cross-Cutting Harm II: Akathisia and the Suicide Link

1.11.1 The Clinical Syndrome of Akathisia

Akathisia is a drug-induced syndrome of subjective inner restlessness and an overwhelming compulsion to move, caused by dopamine D2 blockade in the basal ganglia and mesolimbic system. It is among the most distressing of all pharmacological adverse effects.

Patients describe akathisia in terms that the clinical literature has historically struggled to represent adequately: a feeling of electric current running through the body; an inability to stop moving no matter how exhausted; an inner torture that makes stillness physically impossible; a state that is simultaneously agitation, pain, and despair. The subjective experience of severe akathisia has been described by patients as worse than the psychotic symptoms that led to their admission and treatment.

Akathisia is distinguished from anxiety (which it superficially resembles) by its motor component — the patient must move, pace, or shift weight continuously — and from psychotic agitation by its drug-onset timing and its response to dose reduction or anticholinergic agents. These distinctions are frequently missed in clinical practice.

1.11.2 Misdiagnosis: The Loop That Amplifies the Harm

Akathisia is commonly misidentified by treating clinicians as one of the following, each of which leads to a response that worsens rather than resolves the syndrome:

- **Worsening psychosis:** The restlessness and agitation of akathisia visually resembles psychotic agitation. The clinical response to apparent psychotic agitation is antipsychotic dose increase — which worsens akathisia. The patient's worsening presentation is then taken as evidence of more severe illness, potentially triggering CTO escalation rather than dose reduction;
- **Anxiety disorder:** Akathisia's subjective distress component leads to benzodiazepine prescription, adding polypharmacy without addressing the cause;
- **Personality disorder or treatment resistance:** Patients who report the subjective distress of akathisia — particularly those on CTOs, who have limited power within the clinical relationship — may have their reports attributed to personality pathology or dismissed as manipulative rather than investigated as a drug adverse effect.

1.11.3 The Suicide Risk

Akathisia carries a documented and clinically significant association with suicidal ideation and completed suicide. Van Putten and Marder (1987), in the foundational clinical literature on this subject, documented that akathisia-related subjective distress was among the most powerful predictors of medication non-compliance and associated deterioration in schizophrenia.

Healy et al. (2006) reviewed the evidence linking antipsychotic-induced akathisia specifically to suicidal acts, concluding that the subjective torture of severe akathisia is capable of precipitating suicidal crisis in individuals who were not suicidal before drug initiation. This finding carries a profound clinical and ethical implication: the drug prescribed to reduce distress and suicide risk in acute psychosis may, through akathisia, induce a suicidal crisis in people who would not otherwise have been at that risk level.

In a system that compels antipsychotic administration and where akathisia is frequently misdiagnosed and unaddressed, this is not a theoretical concern. It is a systemic risk with no current monitoring or accountability framework.

1.12 Cross-Cutting Harm III: The Life Expectancy Gap

1.12.1 *The Lawrence et al. Findings: Australian Data*

The most methodologically rigorous available estimate of the life expectancy gap for people with serious mental illness in Australia comes from Lawrence et al. (2013), published in the *BMJ*. This study used Western Australian population-based record linkage — linking psychiatric admission records to mortality registries — to calculate all-cause and cause-specific mortality for people with psychiatric diagnoses relative to the age- and sex-matched general population.

Lawrence et al. (2013), *BMJ* — Australian Life Expectancy Data

- People with schizophrenia: **14.5 years shorter** life expectancy for men, **17.5 years shorter** for women, relative to the general population;
- People with affective (mood) disorders: 9–12 years shorter;
- **85% of excess deaths** were from natural causes — not suicide, violence, or accident — meaning the life expectancy gap is overwhelmingly driven by physical illness, not the psychiatric condition per se;
- Leading causes of excess mortality: **cardiovascular disease, respiratory disease, and endocrine/metabolic disease** — precisely the organ systems most adversely affected by antipsychotic adverse effects.

1.12.2 *The Causal Chain*

The convergence between the causes of excess mortality in Lawrence et al. and the documented adverse effect profiles of antipsychotics is not coincidental. It is causally plausible across multiple well-established biological pathways:

- Antipsychotic-induced weight gain → central adiposity → insulin resistance → type 2 diabetes → cardiovascular disease, renal failure, peripheral vascular disease, and premature death;
- Antipsychotic-induced dyslipidaemia → accelerated atherosclerosis → myocardial infarction and stroke;
- Antipsychotic sedation → physical inactivity → compounded metabolic risk, reduced cardiorespiratory fitness, and obesity;
- Antipsychotic-related prolactin elevation → hypo- gonadism → reduced bone mineral density → osteoporosis → fragility fractures → immobility and associated complications;
- Antipsychotic-related QTc prolongation → sudden cardiac death in vulnerable patients, particularly those on multiple QT-prolonging agents.

De Hert et al. (2011) documented that the excess cardiovascular and metabolic

mortality in people with severe mental illness began *increasing* substantially from the 1990s — precisely the decade in which second-generation antipsychotics with their severe metabolic profiles displaced first-generation drugs as the dominant treatment modality. The temporal correlation is consistent with a causal contribution of the drugs themselves to the widening gap.

1.12.3 The Physical Health Monitoring Failure

Clinical guidelines in Australia, including those of the Royal Australian and New Zealand College of Psychiatrists, recommend that patients on antipsychotics receive baseline and ongoing monitoring including: fasting glucose and HbA1c, fasting lipid profile, body weight and BMI, blood pressure, and waist circumference, at baseline and at three-monthly intervals.

The evidence consistently shows that these guidelines are not followed in the majority of cases in community settings. A patient who is on a Community Treatment Order in Victoria and required to attend a community mental health service fortnightly for depot injection typically does not also receive regular metabolic monitoring. The drug is administered; the harms it causes are not tracked.

A patient who is legally compelled to accept antipsychotic drugs is thereby also compelled to accept the physical health consequences of those drugs, without the monitoring infrastructure that is supposed to detect and manage those consequences. The compulsion flows in one direction only.

1.13 Cross-Cutting Harm IV: Cancer Risk

1.13.1 Prolactin-Dependent Cancers

The association between chronic antipsychotic use and breast cancer risk has been the subject of growing epidemiological evidence, driven by the well-established biology of prolactin as a mitogenic hormone for prolactin-receptor-positive breast tissue.

Solmi et al. (2023) conducted a systematic review and meta-analysis of 14 observational studies covering approximately 500,000 patients, finding a statistically significant association between antipsychotic use and breast cancer risk (pooled relative risk approximately 1.3–1.7), with the association substantially stronger for prolactin-elevating antipsychotics (risperidone, haloperidol) than for prolactin-neutral (olanzapine, quetiapine) or prolactin-lowering drugs (aripiprazole).

This evidence is not yet at the level of definitive causal proof from randomised controlled trials — which cannot be conducted for this question for obvious ethical reasons. However: the biological mechanism is well-established; the direction of the association is consistent across 14 independent datasets; a dose-response relationship is supported by the comparative data across prolactin-elevating and non-elevating drugs;

and the association survives adjustment for confounders in the studies that include them. These are the standard criteria used to infer causation from observational data in pharmacoepidemiology.

Given that prolactin-elevating antipsychotics are among the most commonly prescribed depot formulations in Australia's CTO system — risperidone and paliperidone specifically — the potential population-level cancer burden from this association warrants urgent investigation that has not yet been undertaken.

1.14 The NNT–NNH Comparison: Completing Criterion D

Table 2 consolidates the efficacy and harm metrics for antipsychotics, placing the claimed benefits and the documented harms in direct comparison. This comparison provides the basis for the Criterion D assessment: whether benefits substantially and demonstrably outweigh harms.

Table 2: Antipsychotic NNT versus NNH: efficacy and harm in direct comparison

Metric	Value	Source
<i>Efficacy metrics</i>		
NNT: good response (50% PANSS reduction)	8	Leucht et al., 2017
NNT: relapse prevention (1 year)*	3	Leucht et al., 2017
Real-world 18-month discontinuation (CATIE)	74%	Lieberman et al., 2005
<i>Harm metrics</i>		
NNH: weight gain ($\geq 7\%$ body weight), olanzapine, 10 weeks	3	De Hert et al., 2011
NNH: EPS, first-generation antipsychotics	3–5	Leucht et al., 2017
NNH: akathisia (all antipsychotics)	5–8	Leucht et al., 2017
NNH: tardive dyskinesia, 5-year cumulative	≈ 5	Carbon et al., 2018
NNH: sexual dysfunction (risperidone/paliperidone)	2–3	De Hert et al., 2011
Prevalent TD in long-term SGA users	20.7%	Carbon et al., 2018
Life expectancy reduction (schizophrenia, Australia)	14–17 yrs	Lawrence et al., 2013
Proportion of excess deaths from natural causes	85%	Lawrence et al., 2013

*The relapse-prevention NNT of 3 is measured against abrupt antipsychotic discontinuation and is substantially confounded by dopamine supersensitivity psychosis (see Part III). After correction for this confound (Moncrieff et al., 2020), the true relapse-prevention NNT is estimated to be considerably higher.

NNT = Number Needed to Treat; NNH = Number Needed to Harm; EPS = Extrapyramidal Side Effects; TD = Tardive Dyskinesia; SGA = Second-Generation Antipsychotic; PANSS = Positive and Negative Syndrome Scale.

The table reveals the following:

- The NNH for several serious, documented harms (weight gain, sexual dysfunction, EPS) is comparable to or lower than the NNT for a good therapeutic response (8) — meaning that for every patient who achieves a good response, at least one or two patients will experience a serious adverse effect;

- The relapse-prevention NNT (3), the metric most cited to justify long-term compulsory maintenance, is confounded by supersensitivity psychosis and likely represents a significant overestimate of the true drug-specific benefit. After correction, the NNT and NNH for common serious harms would be comparable or worse;
- 74% of patients in the most rigorous independent trial discontinued treatment within 18 months — meaning the theoretical long-term benefits of antipsychotic maintenance are not realised in the majority of patients in real-world conditions;
- For patients maintained long-term on CTOs — the population for whom this entire analysis is directly relevant — the population-level outcome is a 14–17-year reduction in life expectancy driven primarily by the metabolic and cardiovascular harms of the drugs they have been legally compelled to take.

1.15 Part IV: Summary of Propositions

Propositions Established in Part IV

Proposition 1. *Criterion D violated (benefit–harm balance):* *The documented harm profile of antipsychotics — NNH of 3 for significant weight gain; NNH of 3–5 for EPS; NNH of 5 for tardive dyskinesia over five years; 20.7% prevalent TD in long-term users; 14–17-year life expectancy reduction in Australia — is not demonstrably outweighed by the evidence of benefit, particularly once the long-term outcome data and the supersensitivity confound in relapse statistics are accounted for.*

Proposition 2. *The 14–17-year life expectancy gap for people with schizophrenia in Australia is driven predominantly (85%) by natural-cause mortality in the same organ systems — cardiovascular, metabolic, respiratory — most severely affected by antipsychotic adverse effects. This gap widened from the 1990s with the introduction of metabolically toxic second-generation antipsychotics.*

Proposition 3. *Tardive dyskinesia is an irreversible or persistent disability caused directly by antipsychotics in approximately one in five long-term users. When administered under legal compulsion without consent, and producing a permanent disability, this harm falls within the scope of injury for which, in any other medical context, legal accountability would attach. No such accountability exists in Australia’s compulsory psychiatric treatment framework.*

Proposition 4. *The corporate suppression of metabolic harm evidence by Eli Lilly (olanzapine, \$1.4 billion US federal settlement, 2009) and AstraZeneca (quetiapine,*

\$520 million US settlement, 2010) establishes that the evidence base for two of the most widely prescribed antipsychotics in Australia was deliberately distorted by their manufacturers for commercial gain over the period when prescribing patterns and clinical guidelines were being established.

Proposition 5. *Depot antipsychotic formulations, particularly three-monthly paliperidone palmitate, make truly informed, revocable consent structurally and pharmacologically impossible. Their use under Community Treatment Orders constitutes compelled, irrevocable pharmacological intervention lasting a minimum of three months per dose.*

Proposition 6. *Akathisia — a drug-induced syndrome of motor restlessness that is distressing, frequently misdiagnosed, and associated with suicidal crisis — is a systematic harm of antipsychotic treatment for which no monitoring framework exists in Australia’s CTO population.*

Conclusion of Part IV: Antipsychotics, as administered coercively within Australia’s psychiatric framework, fail all four criteria of the evidentiary standard for justified compulsory treatment: the mechanism is unverified (A), the primary outcome measure is confounded (B), long-term functional outcomes are worse under maintained treatment (C), and the documented population-level harm burden is not demonstrably outweighed by benefit (D).

Part V (Stage 3, forthcoming) will demonstrate that non-pharmacological interventions — exercise, structured social support, Open Dialogue, and related approaches — satisfy all four criteria with superior evidence profiles and zero serious adverse events. This will complete the logical proof that coercive antipsychotic administration in Australia cannot be justified on any scientific, ethical, or legal basis.

References

- [Carbon et al.(2018)] Carbon, M., Kane, J. M., Leucht, S., & Correll, C. U. (2018). Tardive dyskinesia risk with first- and second-generation antipsychotics in comparative randomized controlled trials: A meta-analysis. *World Psychiatry*, **17**(3), 330–340. <https://doi.org/10.1002/wps.20579>
- [De Hert et al.(2011)] De Hert, M., Correll, C. U., Bobes, J., Cetkovich-Bakmas, M., Cohen, D., Asai, I., . . . Leucht, S. (2011). Physical illness in patients with severe mental disorders. I. Prevalence, impact of medications and disparities in health

- care. *World Psychiatry*, **10**(1), 52–77. <https://doi.org/10.1002/j.2051-5545.2011.tb00014.x>
- [Healy et al.(2006)] Healy, D., Herxheimer, A., & Menkes, D. B. (2006). Antidepressants and violence: Problems at the interface of medicine and law. *PLOS Medicine*, **3**(9), e372. <https://doi.org/10.1371/journal.pmed.0030372> See also: Van Putten, T., & Marder, S. R. (1987). Behavioral toxicity of antipsychotic drugs. *Journal of Clinical Psychiatry*, **48**(Suppl. 9), 13–19.
- [Lawrence et al.(2013)] Lawrence, D., Hancock, K. J., & Kisely, S. (2013). The gap in life expectancy from preventable physical illness in psychiatric patients in Western Australia: Retrospective analysis of population based registers. *BMJ*, **346**, f2539. <https://doi.org/10.1136/bmj.f2539>
- [Leucht et al.(2017)] Leucht, S., Leucht, C., Huhn, M., Chaimani, A., Mavridis, D., Helfer, B., . . . Davis, J. M. (2017). Sixty years of placebo-controlled antipsychotic drug trials in acute schizophrenia: Systematic review, Bayesian meta-analysis, and meta-regression of efficacy predictors. *American Journal of Psychiatry*, **174**(10), 927–942. <https://doi.org/10.1176/appi.ajp.2017.16121358>
- [Lieberman et al.(2005)] Lieberman, J. A., Stroup, T. S., McEvoy, J. P., Swartz, M. S., Rosenheck, R. A., Perkins, D. O., . . . Hsiao, J. K. (2005). Effectiveness of antipsychotic drugs in patients with chronic schizophrenia (CATIE). *New England Journal of Medicine*, **353**(12), 1209–1223. <https://doi.org/10.1056/NEJMoa051688>
- [Moncrieff et al.(2020)] Moncrieff, J., Gupta, S., & Hopkins, D. (2020). Reviewing the evidence for the role of dopamine supersensitivity in antipsychotic discontinuation outcomes. *Therapeutic Advances in Psychopharmacology*, **10**, 1–14. <https://doi.org/10.1177/2045125320904104>
- [Moore et al.(2014)] Moore, T. J., Glenmullen, J., & Mattison, D. R. (2014). Reports of pathological gambling, hypersexuality, and compulsive eating associated with dopamine receptor agonist drugs. *JAMA Internal Medicine*, **174**(12), 1930–1933. <https://doi.org/10.1001/jamainternmed.2014.5262>
- [Ronaldson et al.(2011)] Ronaldson, K. J., Fitzgerald, P. B., Taylor, A. J., Topliss, D. J., & McNeil, J. J. (2011). A new monitoring protocol for clozapine-induced myocarditis based on an analysis of 75 cases and 94 controls. *Australian & New Zealand Journal of Psychiatry*, **45**(6), 458–465. <https://doi.org/10.3109/00048674.2011.572852>
- [Solmi et al.(2023)] Solmi, M., Miola, A., Croatto, G., Pigato, G., Favaro, A., Fornaro, M., . . . Correll, C. U. (2023). Risk of breast cancer associated with antipsychotic

drug use: A systematic review and meta-analysis. *European Neuropsychopharmacology*, **72**, 61–68. <https://doi.org/10.1016/j.euroneuro.2023.04.003>